Section III

40 marks
Attempt TWO questions from Questions 25–28
Allow about 1 hour and 10 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

If you include diagrams in your answer, ensure that they are clearly labelled.

Question 25 — Transaction Processing Systems (20 marks)
Use a SEPARATE writing booklet.

(a) Distinguish between a *request* and an *acknowledgement* as features of a transaction. In your answer, use an example. **3**

(b) Customers who would like to attend an online concert need to create an account, then login and pay for their tickets. Once payment has been processed, concert management will send customers a link to an e-ticket.

(i) Describe how the data collected in the registration process could be used by management. **3**

(ii) Draw a data flow diagram for this online ticketing system. **4**

(c) An online application allows office workers to report workplace issues such as: **4**

- login password not working
- ordering paper for a printer
- requesting software to be installed.

Justify the *batch* and *real time* processing that would occur in this application.

(d) Explain the use of *mirroring* and *rollback* procedures in maintaining both data security and data integrity. **6**